



Kumkum Lodhi

Roll No.: 123105124

Bachelors Of Technology

Electronics and Communication Engineering

National Institute Of Technology, Kurukshetra

+91-7357369971

rohittwenty06@gmail.com

123105124@nitkkr.ac.in

GitHub Profile

Portfolio Link

EDUCATION

•National Institute Of Technology, Kurukshetra

2023-2027

Btech, Electronics and Communication Engineering

CGPA : 8.089

•MLP International School,Rajpura

2022

Board of Intermediate Education, Haryana

Percentage: 96.2

•MLP Sr. Sec. School, Maheshwari

2020

Board of Secondary Education, Haryana

Percentage: 97

EXPERIENCE

•Undergraduate Research Intern

May-July 2025

National Institute Of Technology, Kurukshetra

- Wrote a Research paper on the topic 'CNN Based Automatic Modulation Classification in Rician fading Channel'.
- Trained a CNN based model to classify six modulation schemes under rician fading channel and realistic Doppler's conditions.

PERSONAL PROJECTS

• CNN Based Automatic Modulation Classification in Rician fading Channel

May-July 2025

Developed a deep learning-based model using CNN for automatic classification of modulation schemes.

- Tools & technologies used: MATLAB, MS Excel, Tensorflow/Keras frameworks, Pandas
- Improved classification accuracy across varying SNR levels using deep learning techniques and achieved highest accuracy of 94% at 18db SNR , 86% at 0db SNR and overall accuracy of 78%.

• Hyperlocal Delivery Delay Predictor

Project Link

Built end-to-end ML pipeline on 44K orders to predict delivery delays — Swiggy/Zomato inspired

- Tools & technologies used: Python, SQL, DAX (Power BI), Machine Learning, TensorFlow, Power BI, Excel, Matplotlib, Seaborn, FastAPI
- Trained XGBoost + LSTM ensemble achieving $R^2 = 0.82$ and RMSE = 3.99 min, a 41% improvement over Linear Regression baseline.
- Applied SHAP explainability to identify that delivery person age contributes 2.51 min impact — the single biggest delay driver, ahead of distance and traffic.
- Engineered India-specific features: festival calendar (Diwali, Holi), monsoon rain flag, and peak-hour windows — increasing model R^2 by +0.09 over base features alone.
- Built dynamic re-prediction engine that refreshes ETA every 2 minutes using live weather API, reducing prediction error on delayed orders by 22%.
- Deployed FastAPI REST endpoint for real-time scoring and built 3-page Power BI dashboard with KPI cards, city heatmap, SHAP feature chart, and model leaderboard

TECHNICAL SKILLS AND INTERESTS

Languages: C++, HTML/CSS, MATLAB, SQL,Python

Developer Tools: Git, GitHub, Kaggle, Microsoft Excel , Power Bi ,FastAPI

Frameworks: TensorFlow, Keras, PyTorch, Pandas, Numpy, SHAP, LSTM

Cloud/Databases: MySQL

Soft Skills: Attention to Detail, Analytical Thinking, Communication Skills, Problem Solving

Coursework: OOPs, Machine Learning, Deep Learning, MATLAB, Computer Networks, Generative AI

Areas of Interest: Data Analysis, Data Visualization, Dashboarding,Conversational AI, LLM applications

POSITIONS OF RESPONSIBILITY

•Volunteer, SHIKSHA Club

2023 - till date

- Educate and Contribute to educational outreach initiatives aimed at underprivileged students by organizing daily classes and educational drives.